JOSEPH BRUNET

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RESEARCH EXPERIENCE

Research Fellow | University College London (UCL) *Oct 2023 – Present*
Open-ended appointment since September 2024; based at the European Synchrotron Radiation Facility (ESRF), Grenoble, France
Funded by the Chan Zuckerberg Initiative

Lead research on multiscale synchrotron X-ray imaging and quantitative analysis of intact human organs, with a particular focus on cardiac architecture.

- Lead UCL's HiP-CT research activities at ESRF, including experimental campaigns, methodological development, and access to beamline infrastructure
- Manage and support UCL researchers and students based at ESRF, including one PDRA and four UCL-funded ESRF staff
- Coordinate collaborations across eight co-investigator organisations and more than 30 international research groups
- Develop and apply large-volume image-analysis methods, including structure-tensor analysis, tractography, segmentation, and digital volume correlation
- Coordinate UCL's local HiP-CT budget and contribute to project planning, reporting, and risk management
- Co-supervise PhD and MSc students and act as their local supervisor during placements at ESRF

Interim Director, Human Organ Atlas Hub (HOAHub) | University College London (UCL) *May 2025 – Jan 2026*
Concurrent leadership appointment held alongside my Research Fellow position

Directed an interdisciplinary UCL-ESRF research infrastructure producing open-access, multiscale 3D imaging datasets of intact human organs using Hierarchical Phase-Contrast Tomography (HiP-CT).

- Provided scientific and strategic leadership for HOAHub and its international HiP-CT programme
- Coordinated operations across ESRF and eight co-investigator organisations to deliver programme milestones
- Managed collaborations with more than 30 international research groups using the HOAHub infrastructure
- Oversaw the production, curation, and dissemination of open-access data through the Human Organ Atlas platform
- Represented HOAHub internationally and contributed to scientific governance, funding coordination, and programme reporting

Postdoctoral Research Fellow | University College London (UCL) *Sep 2021 – Sep 2023*
Based at the European Synchrotron Radiation Facility (ESRF), Grenoble, France
Funded by the Chan Zuckerberg Initiative

Developed and applied HiP-CT for multiscale imaging and quantitative analysis of intact human and animal organs.

- Served as a UCL visiting scientist at ESRF
- Prepared human and animal biological samples for synchrotron tomography
- Performed Hierarchical Phase-Contrast Tomography experiments on intact organs
- Developed a quasi-dynamic experimental setup for in situ imaging of coronary stent deployment within an intact heart

- Developed and applied image-analysis methods, particularly digital volume correlation, to quantify biological tissue behaviour

PhD Researcher | École des Mines de Saint-Étienne, France *Nov 2017 – Mar 2021*
Funded by the European Research Council
Advisors: Pierre Badel and Éric Maire

Thesis: Understanding the mechanisms of aortic dissection: finite element modelling and *in situ* experimentation with X-ray tomography

- Collected and prepared arterial tissues from pigs and rabbits
- Performed tensile and peeling tests to assess tissue elasticity and fracture properties
- Developed a tension-inflation device for mechanical testing within an X-ray tomography system
- Performed in situ 3D imaging of arterial dissection using laboratory microtomography and synchrotron phase-contrast imaging
- Modelled the observed failure mechanisms using the extended finite element method

Research Intern | Medtronic, France *Jul 2017 – Nov 2017*
Experimental and numerical investigation of surgical meshes for abdominal hernia repair.

- Performed uniaxial, biaxial, and indentation tests on different textile types
- Modelled mesh behaviour using a nonlinear, anisotropic, plastic constitutive model

MSc Researcher | École des Mines de Saint-Étienne, France *Jan 2017 – Jul 2017*
Advisor: Pierre Badel

Thesis: Investigation of shear delamination during rupture of arterial medial tissue using a cohesive numerical model

EDUCATION

École des Mines de Saint-Étienne, France *Nov 2017 - March 2021*
Ph.D. in Biomechanics

Paris Descartes University, France *Sep 2016 - Oct 2017*
Master of Science - MS, Biomedical Engineering - Major in Biomechanics

École Nationale Supérieure d'Arts et Métiers, France *Sep 2014 - Oct 2017*
Master's Degree in Mechanical Engineering

RESEARCH FUNDING AND COMPETITIVE AWARDS

- 2026 - 2030 - ANR CardiacTS - **Partner lead for HiP-CT; total award: 418,693€**
- 2022 - ImagingBioPro Network Proof of Concept Award - **PI, £25,000**

Competitive synchrotron beamtime awards (in-kind value)

- 2024–2026 - HOAHub Biomedical Challenge Beamtime (6 awards; 27 shifts) - **Co-investigator, 153,360€**
- 2024–2026 - HOAHub Feasibility Beamtime (2 awards; 2 shifts) - **PI on one award and co-investigator on one award, 11,360€**
- 2023 - HiP-CT long-term proposal MD-1290 (12 beamtimes) - **Co-investigator, ca. 1,000,000€**
- 2020 - ESRF beamtime LS-2904 - **PI, ca. 85,200€**

TRAVEL GRANTS

- 2025 CZI Imaging the Future 2025: Focusing on Biology in Action, San Jose, US **£2,000**
- 2024 CIFAR MacMillan Multiscale Human Spring 2025, Geneva, Switzerland **£1,500**
- 2024 CZI Imaging the Future 2024: Life Illuminated, Austin, US **£2,000**
- Travel Grants for Invited Talks (6 - equivalent to approx. **£3600**) *2018 - 2024*
  Awarded travel grants to present research at international conferences or workshops

OPEN RESEARCH OUTPUTS

Cardiotensor *Lead developer, 2025–Present*
Open-source Python library for structure-tensor orientation analysis and tractography in large 3D cardiac imaging datasets. Published in the *Journal of Open Source Software* (2026); software and documentation openly available.

Human Organ Atlas open datasets *Data contribution and scientific leadership, 2021–Present*
Named contributor on 375+ open-access dataset DOIs, representing hundreds of terabytes of multiscale 3D human-organ imaging data released through the Human Organ Atlas. Led open-data delivery and international working groups as Interim Director of HOAHub.

AWARDS

1st prize poster competition (£300). CiET Mid-term review, University College London. March, 2024

Camille Mathieu Award (5000€). Handitecam, Arts et Mtiers. December, 2017

SUPERVISION

Co-supervision of 4 doctoral researchers, UCL

- **Vaishnavi Sabarigirivasan, MBPhD** *2022 - 2026*
  HiP-CT phenotyping of tetralogy of Fallot, including myocardial architecture and the atrioventricular conduction system
- **Lisa Chestnutt, PhD** *2024 - 2028 (expected)*
  Deep phenotyping of single-ventricle disease using multiscale HiP-CT
- **Olivia Sheil, PhD** *2024 - 2028 (expected)*
  Cardiac tissue biomechanics using HiP-CT and digital volume correlation, including medical-device design and prototyping
- **Akash Sivananthan, PhD** *2025 - 2029 (expected)*
  Image-based multiscale flow modelling of cardiovascular disease using HiP-CT data

Provide scientific supervision in synchrotron experimental design, image reconstruction, quantitative analysis, and publication.

Co-supervisor of 3 MSc students, ESRF/UCL *2021 - 2023* Projects in mechanical testing, biomedical image analysis, and deep learning.

Supervisor of 2 MSc students, 'Ecole des Mines de Saint-'Etienne *2018 - 2020* Industrial and research projects in biomechanics and experimental mechanics.

TEACHING EXPERIENCE

Organiser and instructor, HiP-CT synchrotron imaging training, ESRF *2021 - 2023*
Designed and delivered practical workshops for PhD students and postdoctoral researchers covering sample preparation, synchrotron scan acquisition, image reconstruction and data handling.

Teaching Assistant, École des Mines de Saint-Étienne (140 hours) *2018 - 2020*
Courses: Experimental mechanics, Finite element method

Tutor for students in difficulty, Saint-Exupery middle school, Macon *2015 - 2016*
Courses: Mathematics and Physics

PROFESSIONAL AND INSTITUTIONAL ROLES

Scientific Lead and Organiser, HOAHub Heart Working Group *2023 - Present*
Lead an international multidisciplinary group coordinating cardiac HiP-CT research across clinical, anatomical, imaging and computational teams. Define scientific priorities and coordinate sample selection, synchrotron experiments, data analysis, publications and collaborations across a portfolio of whole-heart studies in health and disease.

UCL representative at ESRF *2023 - Present*
Locally lead and coordinate research activities, provide assistance to new staff for settling in, facilitate collaboration between UCL and ESRF teams, and ensured smooth operational workflows.

Responsible for health and safety compliance, Human Organ Atlas project at ESRF *2021 - 2025*
Responsible for health and safety procedures related to HiP-CT imaging of human organ samples, including sample handling, ethanol/formalin use, biological material management and synchrotron radiation safety requirements.

Guest Lecturer in Engineering Schools *2022 - 2024*
Delivered presentations on current research and academic opportunities in the sciences to undergraduate engineering students. Aims to inspire students to consider careers in academia.

PhD student representative *2018 - 2020*
Board of the Doctoral School of Science, Engineering and Health of the University of Lyon

PhD student representative *2018 - 2020*
Board of SAINBIOSE laboratory (INSERM)

President of the Saint-Étienne Doctoral Students' Club *2018 - 2019*

PUBLICATIONS

1. Zhou, Y., Occhipinti, E., Elvis, B. K., Chen, J., Megas, S., **J. Brunet**, Purzycka, J., Urban, T., Dejea, H., Teichmann, S. A., Clatworthy, M. R., Tafforeau, P., Lee, P. D. & Walsh, C. L. (2026). LoRCA: LoRA Cycle Adaptation for Histology to HiP-CT Translation with DINOv3, *arXiv*. Preprint. doi: doi: 10.48550/arXiv.2608.10002.

2. Cranley, J., Mach, L., Jimba, T., Kanemaru, K., Zhao, J., Perera, S., Zurke, Y.-X., Koplev, S., Barnett, S., Wang, L., Roberts, K., Kapuge, R., Richardson, L., Polanski, K., Semprich, C., Ho, S. Y., Robertus, J. L., Madrova, M., Tegshjargal, B., Miranda, A. M. A., Sarropoulos, I., Cujba, A.-M., Lee, W., Wilbrey-Clark, A., Palmer, J., **J. Brunet**, Chowdhury, R., Prasad, S., Dark, J., Jrgensen, H., Noseda, M. & Teichmann, S. A. (2026). Multimodal Hypersampling of the adult human heart resolves tissue-specific cell states and disease-associated variants, *bioRxiv*. Preprint. doi: 10.64898/2026.06.16.732617.

3. Sharma, A., Evans, L. A. E., Bourne, L. E., Chen, J., Parmenter, A. L., **J. Brunet**, Madi, K., Marussi, S., Pitsillides, A. A., Lee, P. D. & Staines, K. A. (2026). In Situ Profiling of Nanoscale Strains Uncovers Mechano-Architectural Predictors of Aging and Osteoarthritis Emergence, *Advanced Science*. doi: 10.1002/advs.76716.

4. Almehandi, A., K. Y. C. Li, **J. Brunet**, P. D. Lee, R. H. Anderson & A. C. Cook (2026). Unravelling the Enigmatic Morphology of the Atrioventricular Conduction Axis Using Hierarchical Phase-Contrast Tomography (HiP-CT), *Heart Rhythm*. In press. doi: 10.1016/j.hrthm.2026.07.018.

5. Mirone, A., T. Urban, **J. Brunet**, C. L. Walsh, P. D. Lee & P. Tafforeau (2026). Fast Eikonal Phase Retrieval for High-Throughput Beamlines, *Journal of Synchrotron Radiation*, 33(4), pp. 11691179. doi: 10.1107/S1600577526004273.

6. Jain, Y., Walsh, C. L., Yagis, E., Aslani, S., Nandanwar, S., Zhou, Y., Ha, J., Gustilo, K. S., **J. Brunet**, Rahmani, S., Tafforeau, P., Bellier, A., Weber, G. M., Lee, P. D. & Brner, K. (2026). Vasculature segmentation in 3D hierarchical phase-contrast tomography images of human kidneys, *Nature Communications*. doi: 10.1038/s41467-026-74050-8.

7. Dejea, H.†, **J. Brunet**†, T. Urban, C. Berruyer, A. Mirone, C. Jarnias, A. Bellier, C. L. Walsh, P. D. Lee & P. Tafforeau (2026). Evolution of Hierarchical Phase-Contrast Tomography on the European Synchrotron Beamlines BM05 and BM18: A Whole Adult Human Brain Imaging Case Study, *Journal of Synchrotron Radiation*, 33(4), pp. 11801189. doi: 10.1107/S1600577526005503.

8. Sabarigirivasan, V., **J. Brunet**, Dejea, H., Crucean, A., Jegatheeswaran, A., Capelli, C., Pajaziti, E., Urban, T., Purzycka, J., Tafforeau, P., Walsh, C. L., Lee, P. D. & Cook, A. C. (2026). The heterogeneous nature of atrioventricular conduction tissues in tetralogy of Fallot demonstrated by hierarchical phase-contrast tomography, *JTCVS Structural and Endovascular*, 10, 100111. doi: 10.1016/j.xjse.2026.100111.

9. **J. Brunet**, L. Chestnutt, M. Chourrout, H. Dejea, V. Sabarigirivasan, P. D. Lee & A. C. Cook (2026). Cardiotensor: A Python Library for Orientation Analysis and Tractography in 3D Cardiac Imaging, *Journal of Open Source Software*, 11(121), 9720. doi: 10.21105/joss.09720.

10. Sabarigirivasan, V., **J. Brunet**, H. Dejea, A. Crucean, A. Jegatheeswaran, G. Bosi, T. Urban, L. Chestnutt, J. Purzycka, P. Tafforeau, M. K. Friedberg, P. D. Lee & A. C. Cook (2026). Altered Myocardial Structure in Post-natal Tetralogy of Fallot: A Substrate for Interpretation of Ventricular Function and Dysfunction?, *bioRxiv*. Preprint. doi: 10.64898/2026.04.29.721688.

11. Walsh, C. L., **J. Brunet**, D. Stansby, G. Gaisn, Y. Zhou, M. Ackermann, A. Bellier, C. Berruyer, A. Bocciarelli, et al. (2026). The Human Organ Atlas, *Science Advances*, 12(11). doi: 10.1126/sciadv.adz2240.

12. Wanjau, E., M. Chourrout, C. Maffei, Y. Balbastre, A. Keenlyside, **J. Brunet**, A. Sharma, S. Y. Huang, P. Tafforeau, B. Fischl, A. Yendiki, P. D. Lee & C. L. Walsh (2026). Bridging the microstructural gap in human connectomics using hierarchical phase-contrast tomography as a reference for diffusion MRI in the human brain, *bioRxiv*. Preprint. doi: 10.64898/2026.04.02.715729.

13. Chourrout, M., T. Gong, R. Schalek, A. Keenlyside, Y. Balbastre, N. Karlupia, R. A. Gonzales, I. N. Huszar, E. Wanjau, **J. Brunet**, T. Urban, H. Dejea, D. Stansby, K. Gunalan, B. Glickman, E. J. Gaibor, J. Scherick, K.-M. Bintsi, C. Mauri, C. Analoro, S. Ghosh, A. Bellier, B. R. Fischl, J. Augustinack, P. Tafforeau, C. Maffei, P. D. Lee, J. W. Lichtman, A. Yendiki & C. L. Walsh (2026). Hierarchical X-ray microscopy and mesoscopic diffusion MRI in the same brain reveal the human connectome across scales, *bioRxiv*. Preprint. doi: 10.64898/2026.04.02.716198.

14. Lee, J. Y., D. Alblas, A. Szmul, D. Docter, H. Dejea, Y. Dawood, J. Hanemaaijer-van der Veer, A. Bellier, T. Urban, **J. Brunet**, D. Stansby, J. Purzycka, R. Xue, C. L. Walsh, P. D. Lee, P. Tafforeau, R.-J. Oostra, R. C. J. Kanhai, J. Jacob, J. A. M. van der Post, O. Bleker, S. Both, J. A. F. Huirne & B. S. de Bakker (2026). Neuroanatomy of the clitoris, *bioRxiv*. Preprint. doi: 10.64898/2026.03.18.712572.

15. Chourrout, M., A. Keenlyside, E. Wanjau, Y. Balbastre, E. Yagis, **J. Brunet**, D. Stansby, K. Engel, X. Gui, J. Thoennissen, T. Dickscheid, L. Lamalle, A. Bellier, U. Vivekananda, P. Tafforeau, P. D. Lee & C. L. Walsh (2026). The Extremely Brilliant Brain: An Isotropic Microscale Human Brain Dataset, *bioRxiv*. Preprint. doi: 10.64898/2026.01.27.702076.

16. Docter, D., **J. Brunet**, V. Sabarigirivasan, G. Lemmens, B. Tielemans, J. Sporring, H. Dejea, T. Urban, J. Purzycka, R. Gorter, J. Huirne, V. Michels, J. Hanemaaijer-van der Veer, A. Bellier, D. Stansby, M. Ackermann, R. D. Buelow, D. D. Jonigk, J. Jacob, J. Hagoort, S. Verleden, A. Cook, C. Walsh, P. Tafforeau, P. Lee & B. de Bakker (2026). Hierarchical Phase-Contrast Tomography Imaging: Applicability in biomedical research, *bioRxiv*. Preprint. doi: 10.64898/2026.01.15.699614.

17. Michels, V. M. W., Szmul, A., Jacob, J., Dejea, H., de Bakker, B. S., Huirne, J. A. F. & **HOAHub Uterine Consortium** (2026). Revealing the unseen: 3D synchrotron X-ray imaging of uterine vasculature in adenomyosis, *Angiogenesis*, 29(1), 3. doi: 10.1007/s10456-025-10004-w.

18. Zhou, Y., Aslani, S., Javanmardi, Y., **J. Brunet**, Stansby, D., Carroll, S., Bellier, A., Ackermann, M., Tafforeau, P., Lee, P. D. & Walsh, C. L. (2026) Multiscale segmentation using hierarchical phase-contrast tomography and deep learning, *PLOS Computational Biology*. 22(2), p.e1013923. doi: 10.1371/journal.pcbi.1013923.

19. Urban, T., **J. Brunet**, M. Didziokas, M. Chourrout, A. Keenlyside, K. Orsine Murta Dias, A. Mirone, A. Bellier, H. Dejea, L. Lammalle, B. de Bakker, H. Petrushkin, M. Moazen, C. L. Walsh, P. D. Lee & P. Tafforeau (2025). Resolving near-micron scale features within a whole sheep head using Hierarchical Phase-Contrast Tomography, *bioRxiv*. Preprint. doi: 10.64898/2025.12.15.694379.

20. **J. Brunet**, C. L. Walsh, P. Tafforeau, A. C. Cook, K. Engel, A. Bellier, C. Werlein, F. Lnger, J. Jacob, M. Ackermann, D. D. Jonigk, P. D. Lee (2025). Hierarchical Phase-Contrast Tomography: Advancing High-Resolution Imaging of Intact Human Organs, *Grays Anatomy, 43rd edition*. Book chapter.

21. Newham, E., Parmenter, A. L., Chen, J., Disney, C., Snow, T., **J. Brunet**, Vinci, V., Drnec, J., Sherratt, M. J., Hoyland, J. A., Bellier, A., Bay, B. K., Terrill, N., Lee, P. D. & Gupta, H. S. (2025) Multimodal X-ray imaging reveals hierarchical fibre mechanics, *bioRxiv*. Preprint. doi: 10.1101/2025.09.19.677294.

22. Rahmani, S., Jafree, D. J., Lee, P. D., Tafforeau, P., **J. Brunet**, Nandanwar, S., Jacob, J., Bellier, A., Ackermann, M., Jonigk, D. D., Shipley, R. J., Long, D. A. & Walsh, C. L. (2025) Mapping the arterial vascular network in an intact human kidney using hierarchical phase-contrast tomography, *npj Imaging*. 3(1), p.39. doi: 10.1038/s44303-025-00090-2.

23. Mirone, A., **J. Brunet**, Boistel, R., Sowinski, M., Berruyer, C., Payno, H., Boller, E., Paleo, P., Walsh, C. L., Lee, P. D. & Tafforeau, P. (2025) Eikonal phase retrieval: Unleashing the fourth generation sources' potential for enhanced propagation-based tomography on biological samples, *Journal of Synchrotron Radiation*, 32(5). doi: 10.1107/S1600577525005223.

24. Yagis, E., Aslani, S., Jain, Y., Zhou, Y., Rahmani, S., **J. Brunet**, Bellier, A., Werlein, C., Ackermann, M., Jonigk, D., Tafforeau, P., Lee, P. D. & Walsh, C. (2024) Deep learning for 3D vascular segmentation in hierarchical phase contrast tomography: a case study on kidney, *Scientific Reports*, 14(1), p. 27258. doi: 10.1038/s41598-024-77582-5.

25. **J. Brunet**, Walsh, C., Tafforeau, P., Dejea, H., Cook, A., Bellier, A., Engel, K., Jonigk, D. D., Ackermann, M. & Lee, P. D. (2024) Hierarchical phase-contrast tomography: A non-destructive multiscale imaging approach for whole human organs, *Developments in X-Ray Tomography XV*, Vol. 13152, pp. 1315216. SPIE. doi:10.1117/12.3028717.

26. Verleden, S. E., Urban, T., **J. Brunet**, Wen, W., Peeters, D. J. E., Lapperre, T. S., Walsh, C., Tafforeau, P., Dejea, H., Jonigk, D., Ackermann, M., Van Hoorenbeeck, K., Lee, P. D., Jacob, J. & Hendriks, J. M. H. (2024) Imaging three-dimensional airway morphology in congenital pulmonary airway malformation using hierarchical phase-contrast tomography, *American Journal of Respiratory and Critical Care Medicine*. doi: 10.1164/rccm.202403-0509IM.

27. **J. Brunet**, Cook, A. C., Walsh, C. L., Cranley, J., Tafforeau, P., Engel, K., Arthurs, O., Berruyer, C., Burke OLeary, E., Bellier, A., Torii, R., Werlein, C., Jonigk, D. D., Ackermann, M., Dollman, K. & Lee, P. D. (2024) Multidimensional analysis of the adult human heart in health and disease using hierarchical phase-contrast tomography (HiP-CT), *Radiology*, 312(1), e232731. doi: 10.1148/radiol.232731.

28. Xian, R. P., **J. Brunet**, Huang, Y., Wagner, W. L., Lee, P. D., Tafforeau, P. & Walsh, C. L. (2024) A closer look at high-energy X-ray-induced bubble formation during soft tissue imaging, *Journal of Synchrotron Radiation*, 31(3), pp. 566577. doi: 10.1107/S160057752400290X.

29. **J. Brunet**, Walsh, C. L., Wagner, W. L., Bellier, A., Werlein, C., Marussi, S., Jonigk, D. D., Verleden, S. E., Ackermann, M., Lee, P. D. & Tafforeau, P. (2023) Preparation of large biological samples for high-resolution, hierarchical, synchrotron phase-contrast tomography with multimodal imaging compatibility, *Nature Protocols*, 18(5), pp. 14411461. doi: 10.1038/s41596-023-00804-z.

30. Ackermann, M., Tafforeau, P., **J. Brunet**, Kamp, J. C., Werlein, C., Khnel, M. P., Jacob, J. et al. (2023) Comment on: Intrapulmonary shunt and alveolar dead space in a cohort of patients with acute COVID-19 pneumonitis and early recovery, *European Respiratory Journal*, 61(2). doi: 10.1183/13993003.02121-2022.

31. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E., Lane, B. A., Curt, N., Bravin, A., Laroche, N. & Badel, P. (2023) In situ visualization of aortic dissection propagation in notched rabbit aorta using synchrotron X-ray tomography, *Acta Biomaterialia*, 155, pp. 449460. doi: 10.1016/j.actbio.2022.10.060.

32. **J. Brunet**, Pierrat, B. & Badel, P. (2021) A parametric study on factors influencing the onset and propagation of aortic dissection using the extended finite element method, *IEEE Transactions on Biomedical Engineering*, 68(10), pp. 29182929. doi: 10.1109/TBME.2021.3056022.

33. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E., Curt, N. & Badel, P. (2021) A novel method for in vitro 3D imaging of dissecting pressurized arterial segments using X-ray microtomography, *Experimental Mechanics*, 61, pp. 147157. doi: 10.1007/s11340-020-00645-x.

34. **J. Brunet**, Pierrat, B. & Badel, P. (2021) Review of current advances in the mechanical description and quantification of aortic dissection mechanisms, *IEEE Reviews in Biomedical Engineering*, 14, pp. 240255. doi: 10.1109/RBME.2019.2950140.

35. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E. & Badel, P. (2019) A combined experimental-numerical lamellar-scale approach of tensile rupture in arterial medial tissue using X-ray tomography, *Journal of the Mechanical Behavior of Biomedical Materials*, 95, pp. 116123. doi: 10.1016/j.jmbbm.2019.03.028.

KEYNOTE, PLENARY AND INVITED TALKS

1. "HiP-CT: From Whole Human Organs to the Cellular Scale, Towards Non-Destructive 3D Histology" **Invited talk** to be delivered during the **Histology of the Future** session at the **AFH 2027 Congress**, Paris, France. Date: June 17th, 2027

2. "Exploring the Human Heart across Scales with HiP-CT" **Invited scientific contributor** to an immersive VR session on the **ESC TV Stage** at **ESC Congress 2026**, showcasing whole-heart HiP-CT datasets and their potential for cardiac research and clinical translation. Munich, Germany. Date: August 2026

3. "HiP-CT for Multiscale Mapping of Intact Human Organs" **Keynote lecture** delivered at the **ENRICH Health Conference on Multiscale Imaging**, Paul Scherrer Institute, Switzerland. Date: March 17th, 2026

4. "Synchrotron imaging and organology, from Paganinis violin to the mapping of human organs" **Inaugural lecture** delivered at the **25th Annual Meeting of the Francophone Society of Transplantation**, Grenoble, France. Date: December 2nd, 2025

5. "Pushing the limits of HiP-CT: challenges and opportunities in large-volume biological X-ray imaging" Invited talk delivered at the **Hard X-ray imaging of biological soft tissues symposium 2025**, Crick Institute, London, UK. Date: October 13th, 2025

6. "In-Depth Imaging of the Whole Human Adult Heart via Hierarchical Phase-Contrast Tomography" Invitation for oral presentation at **The power of scattering and imaging in understanding and treating disease** conference organized by the LINXS institute, given at Lund University, Lund, Sweden. Date: November 25th, 2024

7. "Overcoming Data Analysis Challenges: Advancing Deep Learning-Based Segmentation in HOAHub Datasets" Invitation for oral presentation at the **Advancing Medical Imaging - Integrating AI and Data Sharing for the Human Organ Atlas Hub** workshop organized by the Center for Quantification of Imaging Data from MAX IV, given at the Technical University of Denmark, Copenhagen, Denmark. Date: November 26th, 2024

8. "Creating The Human Organ Atlas with HiP-CT" Invitation for oral presentation at the **Big Data and Big Science** workshop organized by University of the Witwatersrand, Johannesburg, South Africa. Date: May 8th, 2024

9. "Human Organ Atlas project : High-resolution imaging of whole human organs and application on COVID-19" **Plenary lecture** at the **XXVImes Journes Francophones de Virologie**, Brussels, Belgium. Date: April 11th, 2024

10. "In situ quantification of the mechanical behavior of arteries using synchrotron X-ray tomography" Invitation for oral presentation at the **GDR Matriaux Fibreux - Caractrisation, modlisation et optimisation (GDR CNRS FIBMAT)**, Lyon, France. Date: December 1st, 2023

11. "BM18, the new ESRF-EBS beamline for multiresolution phase-contrast tomography on large samples : past, present and future" Invitation for **keynote lecture** at the "High-resolution imaging in correlative workflows" workshop organized by Zeiss given at the **Technical University of Munich**, Munich, Germany. Date: July 3rd, 2023

12. "X-ray imaging at BM18 - Human Organ atlas project" Seminar given at **Ecole des Mines de Saint-Etienne**, Saint-Etienne, France. Date: July 3rd, 2023

13. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Invitation for oral presentation at the **6th Annual Workshop on Advances in X-ray imaging**, Harwell, UK. Date: June 22nd, 2023

14. "Human Organ Atlas project: Whole Human Organs Imaged in Synchrotrons" Invitation for oral presentation at workshop (**"3D image analysis From Foundation to Vision"**) at the Center for Quantification of Imaging Data from MAX IV (QIM), Copenhagen, Denmark. Date: March 10th, 2023

15. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Invited talk delivered at the ImagingBioPro Network Workshop (Leveraging Next-Gen Multiscale Imaging at Central Facilities for Biomedical Engineering Research) at the **BioMedEng22**, University College London, London. Date: September 9th, 2022

16. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Seminar given at **Witwatersrand University**, Johannesburg, South Africa. Date: September 6th, 2022

17. "Hierarchical imaging of complete human organs, how the EBS changed the game" Oral presentation at the **Science Day Symposium**, ESRF, France. Date: March 21st, 2022

SELECTED ORAL PRESENTATIONS (12 international conferences in total)

1. "Near-cellular to Whole Organ Quantitative Mapping of the Human Heart Using Hierarchical Phase-Contrast Tomography" Oral presentation delivered at the **BioMedEng25**, Glasgow University, Glasgow, UK. Date: September 5th, 2025

2. "Exceptionally detailed Analysis of Whole Human Adult Heart Using Hierarchical Phase- Contrast Tomography" Oral presentation delivered at the **BioMedEng24**, Queen Mary University of London, London, UK. Date: September 7th, 2024

3. "Creating The Human Organ Atlas with HiP-CT" Invitation for oral presentation at the **SPIE Optics + Photonics**, San Diego, United States. Date: 23 August, 2024

4. "Hierarchical Phase-Contrast Tomography for High-Resolution imaging of Ex Vivo Whole Adult Human Heart in Health and Disease" Oral presentation delivered at the **BioMedEng23**, Swansea University, Swansea, UK. Date: September 14th, 2023

5. "In situ quantification of the mechanical behavior of coronary arteries during angioplasty using synchrotron X-ray tomography in intact porcine hearts" Oral presentation delivered at the **BioMedEng22**, University College London, London. Date: September 9th, 2022

6. "High-resolution imaging of the whole human heart using synchrotron x-ray tomography" Oral presentation delivered at the **5th International Conference on Tomography of Materials and Structures (ICTMS)**, Grenoble, France. Date: July 25th, 2022

7. "3D characterization of crack propagation during the onset of a dissection using X-ray microtomography on pressurized aortic segments" Oral presentation delivered at the **26th Congress of the European Society of Biomechanics**, Online. Date: July 13th, 2021

8. "Investigation of notch propagation in an in-vitro dissection model using X-ray microtomography" Oral presentation delivered at the **45th Congress of the Societe de Biomecanique**, Online. Date: October 27th, 2020

9. "A numerical design of the experiment approach to understand aortic dissection onset and propagation" Oral presentation delivered at the **44th Congress of the Societe de Biomecanique**, Poitiers, France. Date: October 29th, 2019

10. "Chairman of the session 'Macro-scale biofluids'" **44th Congress of the Societe de Biomecanique**, University of Poitiers, France. Date: October 28th, 2019

11. "A new approach combining experiment and numerical simulation using a cohesive interface to model tensile failure in arterial medial tissue at the meso-scale" Poster presentation delivered at the **8th World Congress of Biomechanics**, Dublin, Ireland. Date: July 9th, 2018

12. "Characterization and modeling of rupture in arterial medial tissue under tension from in situ experiments with X-ray tomography" Oral presentation delivered at the **15th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering**, Lisbon, Portugal. Date: March 28th, 2018

PROFESSIONAL AFFILIATIONS

Member of European Society of Biomechanics *2020 - Present*
Member of French Society of Biomechanics *2020 - Present*

REFERENCES

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